Sri Lanka Institute of Information Technology

Project Management Tool Report

For

IoT-Enabled Smart bus route monitoring and revenue management system

Project ID: R26-ISE-004

B.Sc. (Hons) in Information Technology Specializing in Information Systems Engineering

Prepared by:

Student ID	Student Name
IT22228208	Gunasekara W S H R
IT22121042	Ahamed M T M
IT22264152	Sachintha H D D
IT22925268	Aththanayaka N D

Table of Contents

1. Executive Summary 4

1.1. Introduction 4

1.2. Objectives 6

1.2.1. Main Objective 6

1.2.2. Specific Objectives 6

1.3. Scope 7

2. Project Matrices and Task Summary 8

2.1. Screenshots of JIRA – Backlog 8

2.1.1. Project Plan and Team Members 8

2.1.2. Task Allocation 9

2.1.3. Project Timeline 10

2.1.4. Detailed View of a Task 11

2.1.5. Dashboard View of the Project Progress 12

3. Recourse and work Management 13

4. Lessons Learned 15

4.1. Success 15

4.2. Challenges 15

4.3. Improvements 17

5. Conclusion 18

Appendices 19

List of Figures

Figure 1: Project Plan & Team members 8

Figure 2:Task Allocation 9

Figure 3:Project Timeline 10

Figure 4:Detailed View of a Task 11

Figure 5:Dashboard View of the Project Progress 12

List of Tables

Table 1-Budget 19

1. Executive Summary

1.1. Introduction

IoT-enabled Smart Bus Route Monitoring and Revenue Management System is a clever platform designed to make public bus operations smoother. The IoT-Enabled Smart Bus Route Monitoring and Revenue Management System improves how buses are monitored, how they are scheduled, how safe they are, and how they earn money. The IoT-Enabled Smart Bus Route Monitoring and Revenue Management System tackles problems in Sri Lankan public buses. These problems include not having buses when traffic is low, managing timetables by hand, buses straying from their planned routes, unsafe driving habits, sudden stops in service, and old ways of collecting permit fees.

The IoT-Enabled Smart Bus Route Monitoring and Revenue Management System blends technology, GPS vehicle tracking, rule-based scheduling, route compliance checks, driver behavior monitoring, and distance-based charging into one platform. In every bus, the IoT-Enabled Smart Bus Route Monitoring and Revenue Management System installs an ESP32 device that carries a GPS module and an IMU sensor. That device records the bus's location, speed, and motion. The IoT-Enabled Smart Bus Route Monitoring and Revenue Management System forwards this data to a communication layer and back-end services. Those services help managers see what is happening in real time and make smart decisions quickly.

One of the distinguishing characteristics of this system is the event-based approach in monitoring. Rather than sending every GPS coordinate, events in operation, such as starting a trip, completion of the trip, deviation from the assigned route, stopping at certain points, dangerous driving situations, and even malfunctions in vehicles, can be monitored and sent to the backend to reduce unnecessary transmission and storage while providing operational data.

There are four functional areas within the system: The Event-Based Bus Monitoring Engine, Policy-Based Scheduling and Time Slot Engine, Distance-Based Revenue Management Engine, and Driver Monitoring and Operations Reporting Mobile Application. This system creates a continuous connection between actual bus operation, transport policy, safety monitoring, scheduling, and financial management. The research paper also specifies the four functional areas as the base for this framework.

The system has a web interface for the use by transport authorities and mobile applications for the use by bus owners and drivers. Authority users will be able to track the buses, manage the routes and schedules, violation management, breakdown management, and creation of operational reports. Bus owners will be able to see their trips and earnings, while drivers will be able to see their schedules, operational alerts, and breakdown reporting.

The project management tool that is used in the project is JIRA for organizing the activities of the project, assigning the responsibilities, tracking the status of the tasks, and maintaining visibility in the entire development process.

1.2. Objectives

1.2.1. Main Objective

The primary objective of the proposed project is to design an IoT-based intelligent bus management system that would combine real-time vehicle tracking, adherence to routes, scheduling, monitoring the behavior of the drivers, and distance-based revenue collection systems to increase the efficiency and effectiveness of public transport buses.

1.2.2. Specific Objectives

The primary goal will be achieved through the accomplishment of the following secondary goals:

- To design an IoT-enabled monitoring system of vehicles by using ESP32, GPS, and IMU sensors to gather bus operations data.
- To design an event-driven monitoring system to detect key events of bus operations, including departure, arrival, halt, route deviation, and breakdowns.
- To design a system for checking route compliance of buses to spot cases of route deviations from authorized routes.
- To design a policy-based schedule engine to classify bus operations according to the Peak, Normal, and Nighttime slots.
- To change bus schedules and redistribute bus trips upon breakdowns and gaps in the schedule of bus operations.
- To design a distance-based revenue management system based on verified distance traveled and appropriate policies.
- To compute incentives for off-peak/night services that qualify for them based on verified bus operations.
- To monitor bus driver behavior by analyzing GPS and IMU data and detect unsafe events including over-speeding, hard braking, hard turning, and abnormal slowing down.
- To provide a web-based dashboard for the officers of transport authorities to manage bus operations.

1.3. Scope

The scope of the project will include the design and development of an integrated software and IoT-enabled platform for the management and monitoring of public buses.

Four functional components can be identified in the system. First, there is the Event-Driven Bus Monitoring Engine. This module will be responsible for receiving telemetry from buses and will support GPS-based monitoring, monitoring of route compliance, route deviations, trip monitoring, monitoring of stops, and alarms.

Second, there is the Policy-Driven Scheduling and Time-Slot Engine. This module will classify bus trips according to their time as Peak, Normal, and Night depending on the predefined transportation policies. In addition, it will facilitate dynamic scheduling, gap detection, scheduling adjustments, and disruption recovery. For the case of the research implementation, bus trips are classified as off-peak trips if they happen between 9:00 PM and 5:00 AM. They can be eligible for incentives depending on certain conditions being met.

Third, there is the Distance-Based Revenue Management Engine. This module will calculate permit charges based on the validated operational distance. Policy multipliers and incentives are also applied through the mechanism.

The fourth element is the Driver Monitoring and Operational Reporting Mobile Application. GPS and IMU data are used to detect predetermined driving events, whereas the mobile application serves the purpose of delivering to the drivers and owners scheduling information and reports, and also alerting.

ESP32 microcontroller, GPS module, and six-axis IMU are used at the vehicle level. Cloud communication is provided by Azure IoT Hub, and for the backend, Java and Spring Boot together with PostgreSQL database are used. React.js framework is used for the authorities' dashboard implementation, and React Native framework is used for the mobile application development.

This project does not deal with production of bus equipment for commercial use, replacement of current transport infrastructure in the country, or any kind of vehicle control. It only deals with software platform and IoT prototype development and operations decision support, monitoring, scheduling, revenue management, and associated reports.

2. Project Matrices and Task Summary

2.1. Screenshots of JIRA – Backlog

2.1.1. Project Plan and Team members

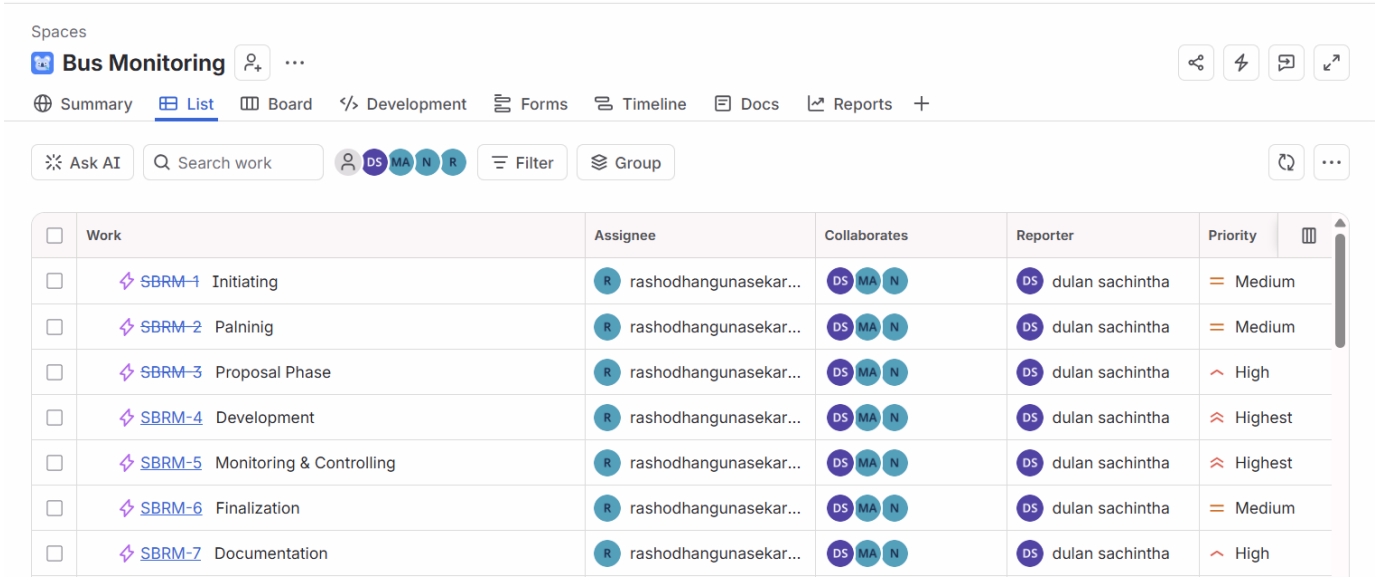


Figure 1: Project Plan & Team members

The JIRA project plan gives an account of the various tasks to be undertaken in the development process of the IoT-Enabled Smart Bus Route Monitoring and Revenue Management System project. Project activities have been grouped according to their respective stages, which include initiation, planning, proposal writing, requirements analysis, system design, development, testing, monitoring and control, documentation, and finally, finalization.

Tasks are assigned to different project team members, which include Rashodhan Gunasekara (Project Team Leader), Dulan Sachintha, Niranga Dineth, and Miflal Ahamed, promoting collaboration, shared responsibility, and coordinated project development.

JIRA task view includes information about task name, priority, status, assignee, and due date. This will enable the project team to track and follow the progress of activities in order to find out what should be paid attention to, while accountability will be achieved due to the allocation of different tasks to specific team members.

2.1.2. Task Allocation

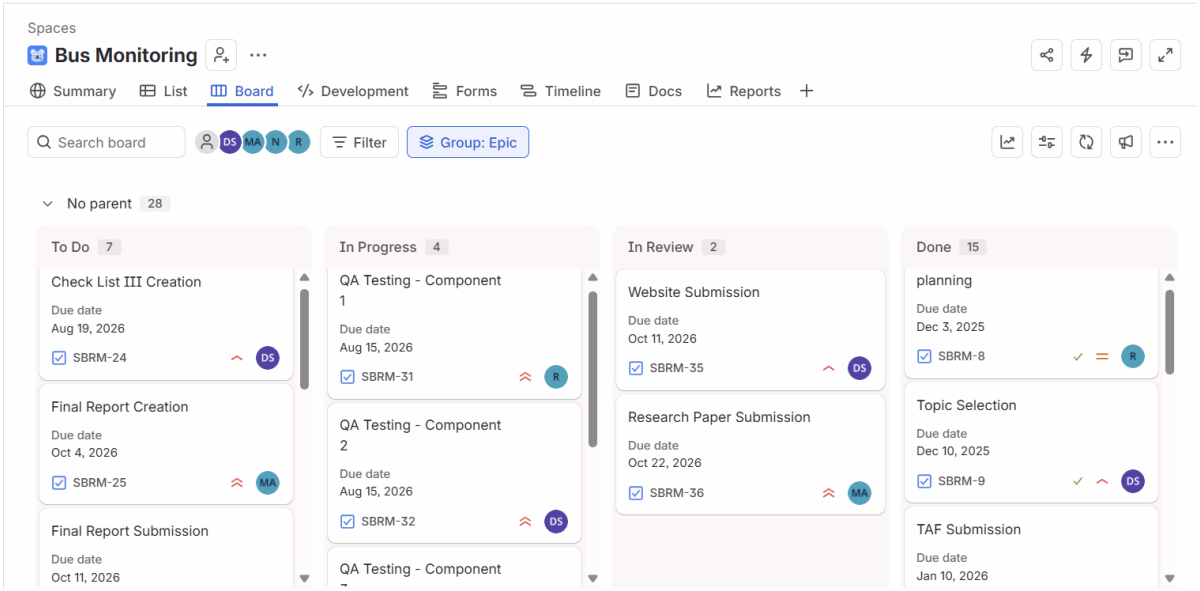


Figure 2:Task Allocation

The board is used in the project to depict the process of execution of project activities. The activities are classified into four stages of workflow, which include: To Do, In Progress, In Review, and Done.

Activities that are to be performed at a later stage and included in the To Do stage include Checklist 3 Creation, Final Report (Overall) Creation, and Final Report.

The activities included in the In Progress workflow include Quality Assurance activities for the three components of the project, that is QA Testing for Component 1, Component 2, and Component 3.

In the In Review stage, there are activities such as Website Submission and Research Paper Submission.

Academic milestones that have been achieved and included in the Done stage include Topic Selection, TAF Submission, and Charter Submission.

2.1.3. Project Timeline

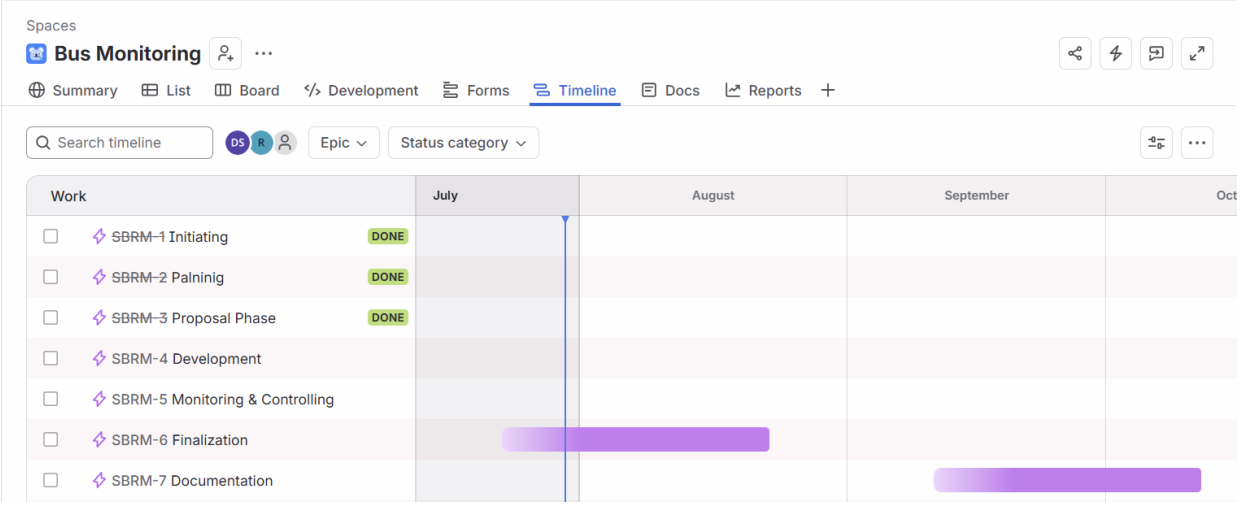


Figure 3:Project Timeline

The JIRA Timeline is a chronological representation of the project's key events. The project timeline starts from early academic events like Topic Selection, TAF Submission, and Charter Submission, and goes through all project stages: proposal writing, requirement analysis, design, implementation, testing, documentation, and finally submission.

The timeline gives the team an understanding of how long project activities take place and in what order they happen. In addition, the timeline gives information about activity dependencies and some critical deadlines like Website Submission and Research Paper Submission.

With the help of a timeline, the team can trace whether the activities have been done on time or not.

2.1.4. Detailed View of a Task

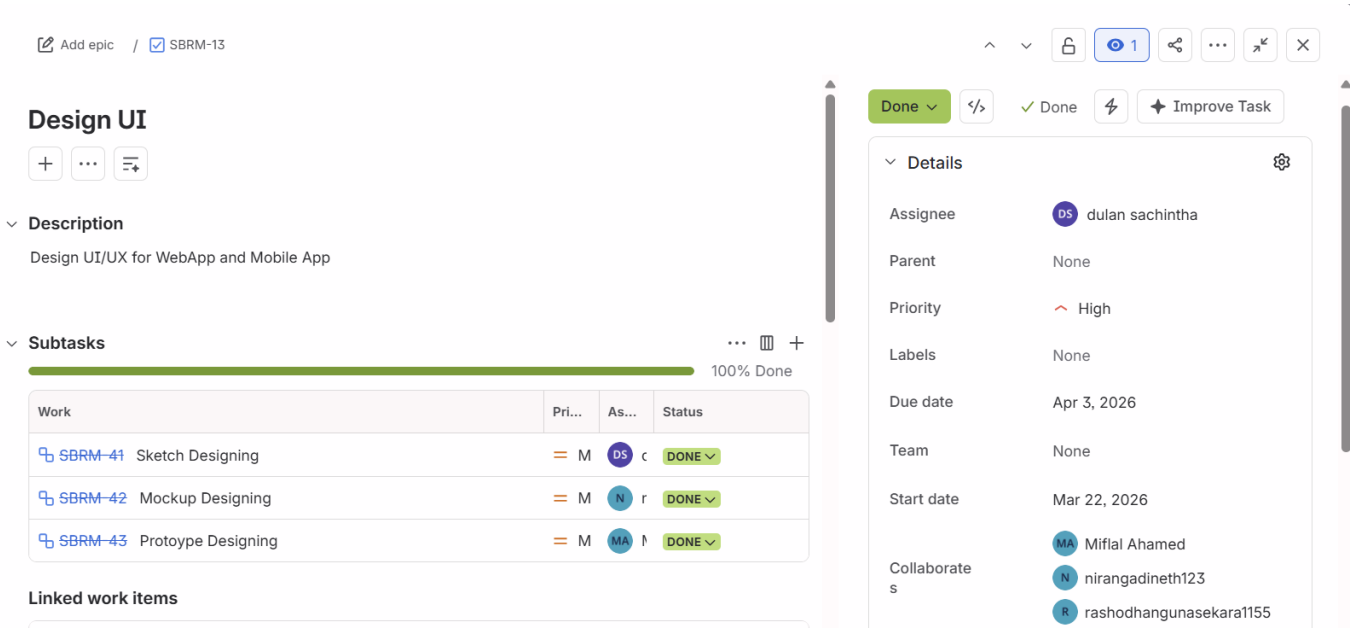


Figure 4:Detailed View of a Task

JIRA is a project management tool that helps teams of any size deliver value faster while minimizing bureaucracy and overhead. This screenshot reflects the detailed view of one particular task (Jira Issue) within the IoT-Enabled Smart Bus Route Monitoring and Revenue Management System. The task has relevant information such as the task's description, attachments, subtasks, assignee, collaborators, task's priority, and schedule. The person in charge of this specific task is Dulan Sachintha. He has received the assistance of his colleagues: Rashodhan Gunasekara (team leader), Niranga Dineth, and Miflal Ahamed. The task's subtasks highlight the collaborative nature of the work and the work needed to address the task and ensure that the team tracks progress on a granular level. The detailed view of the task ensures that the team can manage their responsibilities efficiently, track the progress of their tasks, facilitate teamwork and collaboration, and maintain accountability and documentation throughout the development lifecycle.

2.1.5. Dashboard View of the Project Progress

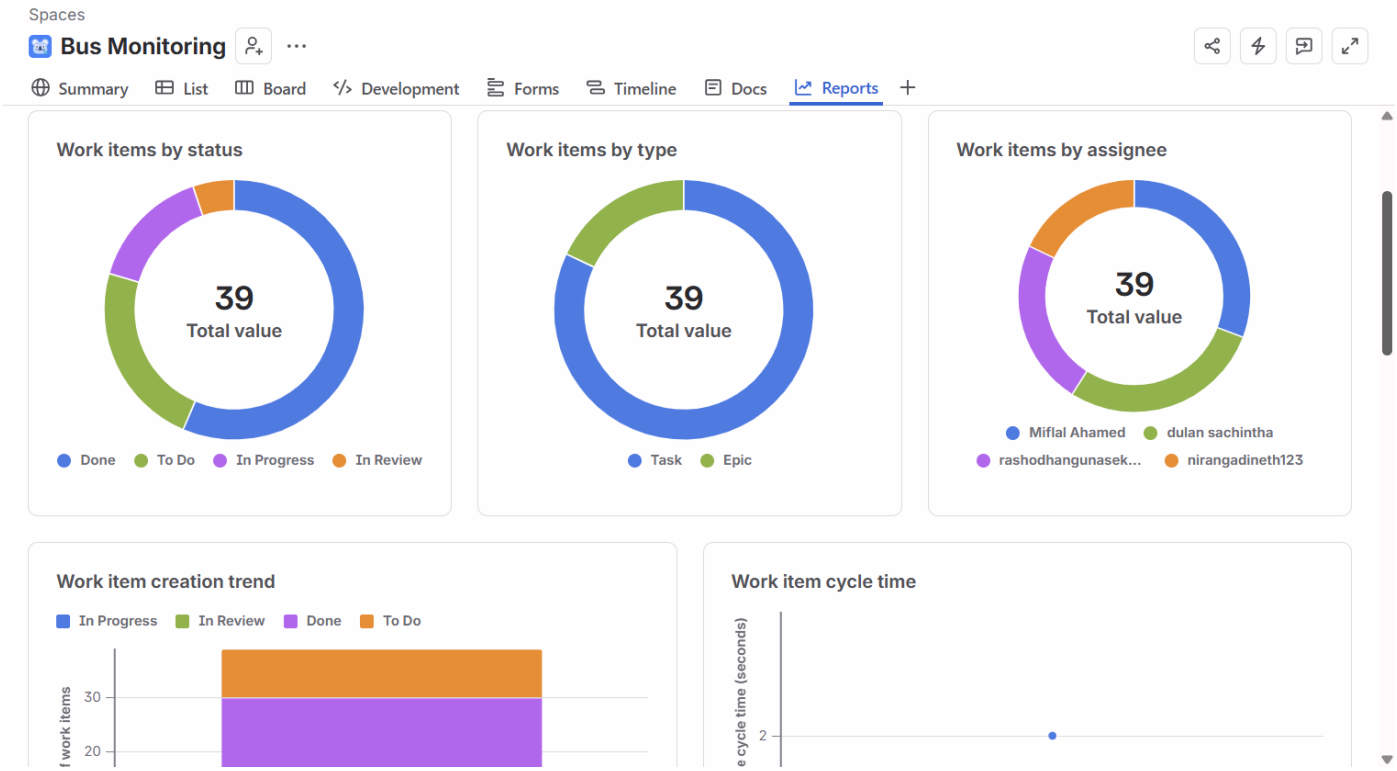


Figure 5:Dashboard View of the Project Progress

Overall project status is represented in the JIRA dashboard. Information about projects is presented using various graphics, which include issue statistics, charts, assigned tasks, recently modified issues, and activity information.

Through the use of the dashboard, the team can see how many tasks have been completed, which tasks are being undertaken, reviewed tasks, and tasks that still need to be completed. It also ensures that project activities are centrally monitored and project phases can be tracked easily by the team.

The dashboard serves as one point where the entire process of the project can be monitored.

3. Recourse and work Management

The development of the IoT-Enabled Smart Bus Route Monitoring and Revenue Management System is done by a team of four members: Rashodhan Gunasekara (Team Leader), Dulan Sachintha, Niranga Dineth, and Miflal Ahamed. Each member is assigned certain tasks related to their specific function within the scope of work of the project.

Several types of technology resources are required to implement the project, namely an ESP32 microcontroller, a GPS module, an IMU sensor, a developer computer, and a vehicle IoT prototype. Software technologies used to develop the system include Java and Spring Boot for back-end development, React.js for the authority interface, React Native for the mobile app, and PostgreSQL for the database. Azure IoT Hub acts as cloud-based communication technology for IoT devices, and other tools are used to develop IoT and test API (Arduino IDE, GitHub, Postman).

JIRA is used to organize and monitor project work. Tasks are divided into manageable activities and assigned to team members with relevant priorities and deadlines. The Kanban board provides a visual workflow for monitoring task progress, while the Timeline supports project scheduling and deadline management.
The project follows an iterative development process in which development, integration, testing, and documentation activities are performed throughout the project lifecycle. Regular communication between team members helps coordinate the four main system components and address integration issues.

The use of centralized project management and development tools improves transparency and allows the team to maintain a structured record of project activities, responsibilities, and progress.

JIRA is used to plan and control the project work. Activities of the project are planned and distributed among members of the team according to their priorities and deadlines. The Kanban board is used to control task flow visually. The Timeline is useful in planning and managing deadlines.

Project is based on the iterative approach, in which all activities of development, integration, testing, and documentation take place during the whole project lifecycle. Communication among

the team members is useful for coordinating the four main system components and solving integration problems.

Centralized management and development tools make the team able to keep control over all project activities and their results.

4. Lessons Learned

4.1. Success

There were quite a few significant achievements throughout the course of development of this project. The implementation of JIRA made the whole process easier, since it helped to organize all the project tasks and responsibilities, monitor their progress, and ensure the visibility of the project itself.

The segmentation of the system into four main functional parts allowed the team to deal with the complexity of the project. This way, there is vehicle monitoring, scheduling based on policies, revenue management, and driver monitoring functionality.

Integration of IoT hardware with cloud communication and software is yet another significant achievement.
The project also successfully developed mechanisms for route compliance, time-slot classification, revenue calculation based on distance and monitoring of driver behaviour. The research evaluation stated that 96.2% accuracy of time-slot classification, 95.1% policy-rule enforcement accuracy and an overall of 92.6% detection accuracy of driver behaviour.

The dynamic breakdown recovery concept was also an important system capability as it allows operational disruptions to be addressed by the current bus availability rather than relying entirely on the manual scheduling process.

4.2. Challenges

Several challenges were encountered during the development of the system. One of the challenges was the integration of multiple technologies and components, including IoT hardware, GPS and IMU sensors, cloud communication, backend, database services, web application, and mobile application.

Another challenge was the GPS-based verification of the route taken by the buses. Since the GPS data may contain errors, the system requires map-matching and route-corridor validation to decide whether a bus sticks to its route. Specifically, the system detects deviation if the bus travels outside the route corridor for more than 30 seconds.

The dynamic nature of scheduling also posed a significant challenge. The system should be able to adjust schedules due to delays and breakdowns by identifying affected trips and finding appropriate rescheduling opportunities while ensuring route services.

Another challenge was the integration of the revenue calculation mechanism. The system requires validated data about the distance covered to calculate expenses and revenues accurately. This poses a great challenge since the system involves permit fee and incentive calculations.

Team organization and time management also posed a significant challenge. The project involved multiple components that had to be developed and integrated.

4.3. Improvements

The project development process can be improved by allocating more time for early planning, integration, and testing. More detailed task breakdowns in JIRA can help the team to identify dependencies between IoT, backend, frontend, mobile, and database activities.

Regular technical integration meetings can also reduce the issues that occur when individual components are developed separately. Continuous integration testing can be performed instead of waiting until the final development stage.

Automated testing can further improve the system reliability, particularly in scheduling rules, revenue calculations, route-compliance logic, and API functionality.

The project can also improve its testing process by using larger real-world datasets and field scenarios. The current research evaluation includes controlled and simulated scenarios across tracking, scheduling, revenue, and driver behaviour functions.

Future developments can also consider the research paper's identified future directions, including edge-AI-based driver behaviour classification, passenger-density estimation, and multi-route field evaluations.

5. Conclusion

The IoT-Enabled Smart Bus Route Monitoring and Revenue Management System demonstrates how IoT, cloud computing, intelligent scheduling, route monitoring, driver safety monitoring, and revenue management can be integrated into a single public transportation management platform.

The project addresses operational problems associated with manual bus monitoring, route deviations, service disruptions, off-peak service availability, unsafe driving behaviour, and conventional revenue management. The integration of ESP32-based IoT devices, GPS and IMU sensors, Azure IoT Hub, Spring Boot, PostgreSQL, React.js, and React Native provides the technical foundation for the proposed solution.

JIRA played an important role in managing the project by providing a centralized platform for task allocation, progress monitoring, scheduling, collaboration, and reporting. The use of Epics, tasks, subtasks, Kanban workflow, Timeline, and Dashboard views enabled the team to manage the project activities in a structured manner.

The project also provided valuable experience in integrating hardware and software components, managing a multi-component system, coordinating team responsibilities, testing complex business rules, and handling project deadlines.

Overall, the project provides a structured foundation for intelligent bus management by connecting actual vehicle operations with scheduling policies, route compliance, driver safety, and financial management. The evaluated prototype demonstrates the feasibility of the proposed integrated approach and provides opportunities for further development and real-world field evaluation.

Appendices

DESCRIPTION	AMOUNT (LKR)	TOTAL (LKR)
SOFTWARE		
React.js	6 000	
Spring Boot	10 000	
React native	10 000	
C++	7 000	
PostgreSQL	9 000	
Visual Code	5 000	
Software Cost		47 000
HARDWARE		
Router	6 000	
2 pc	40 000	
IoT Device	9 000	
Total Hardware Cost		55 000
NETWORK		
Data transfer and API usage	8 000	
Hosting and server deployment	7 000	
Total Networking Cost		15 000
MAINTENCE		
Client training and technical support	6 000	
Warranty and support services	10 000	
Security software and monitoring	4 000	
System maintenance agreement	5 000	
Total Maintenance Cost		25 000
Total Project Cost		**142 000**

Table 1-Budget